Dongyoon Hwang

Ph.D. Candidate in Artificial Intelligence

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Summary

I am interested in building general-purpose robots that can reliably perceive, reason, and act in the unstructured physical world. My research focuses on (i) grounding robot foundation models in explicit 3D geometry to connect perception with action, and (ii) developing scalable and data-efficient reinforcement learning methods that enable robots to acquire robust control policies through interaction with their environments. I am currently a Ph.D. candidate at the DAVIAN Lab, KAIST AI, advised by Prof. Jaegul Choo. Previously, I was a research intern at Krafton AI and Naver Labs Europe, and I will join Microsoft Research Asia in Tokyo as a research intern in December 2026.

Field of Interest

My research interests include, but are not limited to:

- Robot foundation models that perceive and act in metric 3D space [P1, P3, C3]
- Scalable and data-efficient reinforcement learning [P2, C2, C5]
- Visual representation learning for embodied agents [C4, C7, C9]

Education

Korea Advanced Institute of Science and Technology (KAIST) — *Seongnam, Republic of Korea*

Integrated M.S./Ph.D. in Artificial Intelligence (Advisor: Prof. Jaegul Choo, DAVIAN Lab) — *Sep. 2021 - Present*

Korea University — *Seoul, Republic of Korea*

B.S. in Computer Science — *Mar. 2015 - Aug. 2021*

Work Experience

Microsoft Research Asia — *Tokyo, Japan*

Research Intern — *Starting Dec. 2026*

- Data-efficient reinforcement learning for real-world adaptation of robot foundation models

Krafton AI — *Seoul, Republic of Korea*

Research Intern — *Nov. 2024 - Nov. 2025*

- Built an AI co-playable character (CPC) for PUBG Ally on top of LLM/VLM-based agent systems. [demo]
- Developed an LLM-based chess-reasoning agent post-trained with reinforcement learning [W1].

Naver Labs Europe — *Grenoble, France*

Research Intern — *Nov. 2022 - Feb. 2023*

- Studied representation learning for reinforcement-learning-based recommender systems [C10].

Selected Publications

[P1] **PointmapVLA: Scaling Robot Learning with Robot-Centric Pointmaps**
Dongyoon Hwang*, Dongjin Kim*, Byungkun Lee*, Kyungmin Lee, Dohyun Lee, Hojoon Lee, Minho Park, and Jaegul Choo
Preprint, 2026 [paper] [project page]

[P2] **FlashDexRetarget: Accelerating Dexterous Manipulation Data Generation through Multi-Motion Retargeting**
Kyungmin Lee*, Sibeen Kim*, **Dongyoon Hwang***, Yoonsang Oh*, Donghu Kim, Youngdo Lee, I Made Aswin Nahrendra, Jaegul Choo, and Hojoon Lee
Preprint, 2026 [paper] [project page]

[C3] **3D HAMSTER: Bridging Planning and Control in Hierarchical Vision Language Action Models through 3D Trajectory Guidance**

Dongyoon Hwang*, Byungkun Lee*, Dongjin Kim*, Hyojin Jang, Hoiyeong Jin, Jueun Mun, Minho Park, Hojoon Lee, Hyunseung Kim, and Jaegul Choo
International Conference on Intelligent Robots and Systems (IROS), 2026 [paper] [project page]

[C5] **SimBa: Simplicity Bias for Scaling Up Parameters in Deep Reinforcement Learning**
Hojoon Lee*, **Dongyoon Hwang***, Donghu Kim, Hyunseung Kim, Jun Jet Tai, Kaushik Subramanian, Peter R. Wurman, Jaegul Choo, Peter Stone, and Takuma Seno
International Conference on Learning Representations (ICLR), ***spotlight****, 2025* [paper]

Publications

(* equal contribution)

Robotics

[P1] **PointmapVLA: Scaling Robot Learning with Robot-Centric Pointmaps**
Dongyoon Hwang*, Dongjin Kim*, Byungkun Lee*, Kyungmin Lee, Dohyun Lee, Hojoon Lee, Minho Park, and Jaegul Choo
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Preprint, 2026 [paper] [project page]

[P3] **See like a Robot: Robot-Centric Pointmaps for VLA Models**
Byungkun Lee*, **Dongyoon Hwang***, Dongjin Kim, Hojoon Lee, Hyunseung Kim, Jaegul Choo, and Minho Park
Preprint, 2026 [paper] [project page]

[C1] **An Agentic Framework for Proactive Robots with Procedural Assistance Memory**
Dohyun Lee*, Minho Park*, Jeonghoon Park, **Dongyoon Hwang**, Junha Hyung, Sungwon Hwang, Byungkun Lee, and Jaegul Choo
Conference on Robot Learning (CoRL), 2026 [paper] [project page]

[C2] **PHUMA: Physically Reliable Humanoid Locomotion Dataset**
Kyungmin Lee*, Sibeen Kim*, Youngdo Lee, Minho Park, Hyunseung Kim, **Dongyoon Hwang**, Donghu Kim, Hojoon Lee, and Jaegul Choo
Conference on Robot Learning (CoRL), ***spotlight****, 2026* [paper] [project page]

[C3] **3D HAMSTER: Bridging Planning and Control in Hierarchical Vision Language Action Models through 3D Trajectory Guidance**
Dongyoon Hwang*, Byungkun Lee*, Dongjin Kim*, Hyojin Jang, Hoiyeong Jin, Jueun Mun, Minho Park, Hojoon Lee, Hyunseung Kim, and Jaegul Choo
International Conference on Intelligent Robots and Systems (IROS), 2026 [paper] [project page]

[C4] **Adapting Pretrained ViTs with Convolution Injector for Visuo-Motor Control**
Dongyoon Hwang*, Byungkun Lee*, Hojoon Lee, Hyunseung Kim, and Jaegul Choo
International Conference on Machine Learning (ICML), 2024 [paper]

Reinforcement Learning

[C5] **SimBa: Simplicity Bias for Scaling Up Parameters in Deep Reinforcement Learning**
Hojoon Lee*, **Dongyoon Hwang***, Donghu Kim, Hyunseung Kim, Jun Jet Tai, Kaushik Subramanian, Peter R. Wurman, Jaegul Choo, Peter Stone, and Takuma Seno
International Conference on Learning Representations (ICLR), ***spotlight****, 2025* [paper]

[C6] **Do's and Don'ts: Learning Desirable Skills with Instruction Videos**
Hyunseung Kim, Byungkun Lee, Hojoon Lee, **Dongyoon Hwang**, Donghu Kim, and Jaegul Choo
Neural Information Processing Systems (NeurIPS), 2024 [paper]

[C7] **Investigating Pre-Training Objectives for Generalization in Vision-Based Reinforcement Learning**
Donghu Kim*, Hojoon Lee*, Kyungmin Lee*, **Dongyoon Hwang**, and Jaegul Choo
International Conference on Machine Learning (ICML), 2024 [paper]

[C8] **Learning to Discover Skills through Guidance**
Hyunseung Kim*, Byungkun Lee*, Hojoon Lee, **Dongyoon Hwang**, Sejik Park, Kyushik Min, and Jaegul Choo
Neural Information Processing Systems (NeurIPS), 2023 [paper]

[C9] **On the Importance of Feature Decorrelation for Unsupervised Representation Learning in Reinforcement Learning**
Hojoon Lee, Koanho Lee, **Dongyoon Hwang**, Hyunho Lee, Byungkun Lee, and Jaegul Choo
International Conference on Machine Learning (ICML), 2023 [paper]

Others

[W1] **Can Large Language Models Develop Strategic Reasoning? Post-training Insights from Learning Chess**
Dongyoon Hwang*, Hojoon Lee*, Jaegul Choo, Dongmin Park, and Jongho Park
Conference on Language Modeling (CoLM), SCALR workshop, 2025 [paper]

[C10] **SARDINE: Simulator for Automated Recommendation in Dynamic and Interactive Environments**
Romain Deffayet, Thibaut Thonet, **Dongyoon Hwang**, Vassilissa Lehoux, Jean-Michel Renders, and Maarten de Rijke
ACM Transactions on Recommender Systems (TORS), 2024 [paper]

[C11] **Towards Validating Long-Term User Feedbacks in Interactive Recommendation Systems**
Hojoon Lee, **Dongyoon Hwang**, Kyushik Min, and Jaegul Choo
ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), short paper, ***best short paper honorable mention****, 2022* [paper]

[C12] **DraftRec: Personalized Draft Recommendation for Winning in Multi-Player Online Battle Arena Games**
Hojoon Lee*, **Dongyoon Hwang***, Hyunseung Kim, Byungkun Lee, and Jaegul Choo
ACM Web Conference (WWW), 2022 [paper]

[P4] **MOI-Mixer: Improving MLP-Mixer with Multi Order Interactions in Sequential Recommendation**
Hojoon Lee, **Dongyoon Hwang**, Sunghwan Hong, Changyeon Kim, Seungryong Kim, and Jaegul Choo
arXiv preprint, 2021 [paper]

Honors & Awards

2025 **Spotlight Presentation (top 5%),** ***SimBa*** **[C5]**, ICLR
2022 **Best Short Paper Honorable Mention Award [C11]**, ACM SIGIR
2021 **Graduation Project Award, Silver Prize (2,000 USD)**, Korea University